

Vutukuri Sathvik

FULL-STACK ENGINEER · BACKEND & DISTRIBUTED SYSTEMS

11+ years as a **full-stack engineer** building complete products end to end — from large-scale **backend and distributed systems** in Java (high-throughput messaging, distributed scheduling, decisioning infrastructure) to the **React** web portals and dashboards on top. Deepest on the systems side, owning architecture through **production delivery** at scale (platforms serving 1B+ requests/day), and comfortable across the full stack.

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EXPERIENCE

Member of Technical Staff I · Backend Architect

PayPal

Bengaluru, IN

Apr 2022 – Present

🧠 KAIROS — Communication Decision Platform

- Designed core components of **KAIROS**, PayPal's Communication Decision Platform — the centralized layer that governs what every consumer is communicated, when, and on which channel — designed, delivered, and running in production.
- Designed the **Intent Management System (IMS)** — the ingestion and management layer for decision intents flowing into the platform from producing systems.
- Architected the **Deferred-Intent Scheduler** — a durable, leaderless distributed timer on **Google Bigtable + Pulsar** sized for ~25–30M deferred intents/day with bursts up to 2M due at once. Used per-row atomic **compare-and-set (checkAndMutate)** claims for single-fire dispatch and **Kafka consumer-group** hash-partitioning for leaderless work distribution across autoscaling instances.
- Owned the Scheduler's end-to-end low-level design: hash-prefixed row-key schema for even node distribution, scan-claim-dispatch pipeline, cell-timestamp-based garbage collection, failure recovery, and at-least-once dispatch with idempotent re-evaluation.
- Evaluated alternative Bigtable data models (row-per-intent vs wide-row bucketing) with DBA architects, selecting the design that preserved write parallelism at burst scale.
- Designed part of the **Ledger** component and defined the ingestion/dispatch contracts integrating the Scheduler with the Decision Engine's trigger pipeline.

⚡ PayPal Communications — Comms Platform

- Architected and own the end-to-end backend platform powering all PayPal communications — sending, scheduling, and delivering across **Push, Notification Center, Critical Alert & Silent Push** for PayPal, Venmo & Xoom tenants.
- Driving the **communications platform revamp** — redesigning service architecture for scalability, multi-region support, and integration with the centralized decisioning layer.
- Designed and built **Beacon** — the production web portal for PayPal Communications used company-wide to schedule and manage communications; owned it end-to-end (React frontend).
- Made the comms platform a **producer of intent objects** so every send/suppress/defer decision is centrally evaluated against eligibility, ML scoring, cap enforcement & quiet hours.

📄 Server-Driven UI (SDUI) Platform

- Led backend for PayPal's in-house **Server-Driven UI platform** — built from scratch and shipped to production, serving **1 billion+ requests/day** and powering core PayPal app & web experiences.
- Architected core SDUI services: version management, page inheritance, schema versioning & database design — enabling consistent UI delivery across iOS, Android & Web.

🏆 PayPal SPOT Awards — Q3 2022 & Q4 2023

Senior Software Engineer

Informatica

Bengaluru, IN

Mar 2020 – Apr 2022

- Architected and built **SecureStats** end-to-end — security-flaw management platform with full backend, frontend, database design & APIs.
- Co-designed **Flaw Management Service & Notification Service** in a microservices architecture.
- Mentored 5 engineers on service design & architecture decisions. **INFASTAR Award Q1 2021.**

SKILLS

LANGUAGES

Java (8–25) JavaScript

SQL

DISTRIBUTED SYSTEMS

System Design

Distributed Scheduling

Event-Driven

Consistency & Idempotency

High Throughput

Fault Tolerance

CONCURRENCY

Virtual Threads

Structured Concurrency

CompletableFuture

Multithreading

BACKEND

Spring Boot GraphQL

REST Microservices

gRPC

MESSAGING & STREAMING

Kafka Pulsar

DATA & STORAGE

Bigtable Oracle

MongoDB

CLOUD & INFRA

GCP GKE Kubernetes

Docker Amazon S3

FRONTEND / FULL-STACK

React JS Redux

JavaScript Tailwind

HTML/CSS

Admin Dashboards

OBSERVABILITY

Datadog Grafana

Prometheus Micrometer

OpenTelemetry

- Senior Software Engineer

Envestnet Yodlee

Architected **Juggernaut** — microservice platform for automated issue resolution on Yodlee PFM. Owned end-to-end backend & frontend design.

Designed observability stack with Spring Boot Actuator, Micrometer, Prometheus & Grafana; microservice infra with Eureka, Zuul & Hystrix.

Yodlee Dream Team Award — Juggernaut R&D (Q2 2018, Q2 2019).
- Bengaluru, IN

Nov 2015 — Mar 2020

SELECTED TECHNICAL HIGHLIGHTS

- ▶ **Distributed scheduling at scale** — designed a durable, leaderless timer on Bigtable + Pulsar for ~25–30M deferred intents/day with 2M-at-once bursts, using per-row atomic compare-and-set for single-fire dispatch without a coordinator.
- ▶ **High-throughput serving** — led the backend for a Server-Driven UI platform serving **1B+ requests/day** across iOS, Android & Web.
- ▶ **Schema & storage design** — evaluated competing Bigtable data models with DBA architects and chose the design that preserved write parallelism under burst load.
- ▶ **Messaging & event-driven systems** — built platforms on Kafka and Pulsar spanning ingestion, scheduling, and dispatch, with idempotent, at-least-once delivery semantics.
- ▶ **Full-stack delivery** — built the React web portals and admin dashboards on top of the backend systems (e.g. Beacon), owning features from data model to UI.
- ▶ **End-to-end ownership** — consistently took systems from problem statement and architecture through low-level design, implementation, and production rollout.

FOUNDATIONS

- DSA OOP
- Design Patterns

EDUCATION

B.Tech — CSE
JNTU Kakinada · GPA 7.9/10
2011 – 2015

AWARDS

- 🏆 **PayPal SPOT Awards**

SDUI · Q3 2022, Q4 2023
- ★ **Informatica INFSTAR**

SecureStats · Q1 2021
- 🏆 **Yodlee Dream Team**

Juggernaut R&D · Q2 2018, Q2 2019
- 🌟 **Yodlee Spot Awards**

Q2/Q3 2016 · Q1 2018 · Q3 2019